

Parts of Speech Tagging in NLP

In simple words, we can say that POS tagging is a task of labelling each word in a sentence with its appropriate part of speech.

↓
Category of words having similar grammatical properties

Applications of POS tagging:

1. Named Entity Recognition
2. Question Answering System
3. Word sense (~~disambiguation~~) disambiguation
4. Chatbots

Bayes Theorem

Dev

Weather	Mood
Sunny	Happy 😊
Rainy	Grumpy 😞

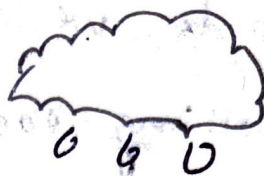
↓

Lalitha

Straight forward,
clear distinction

Adding some complexity

Emission Prob's



Happy
0.8

Grumpy
0.2

Happy
0.4

Grumpy
0.6

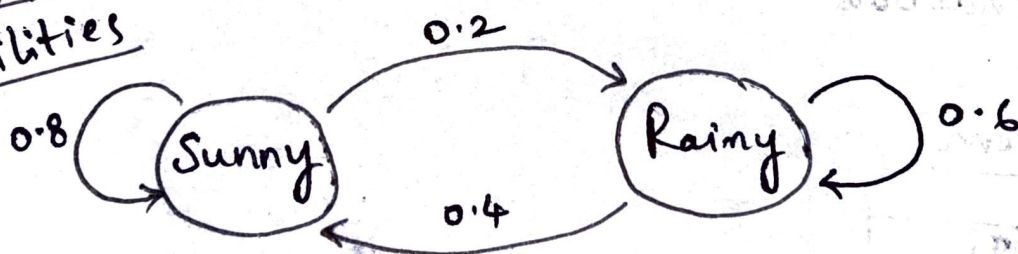
Dev says the following:

Day 1 2 3 4 5 6 7

Happy Grumpy H H G H G

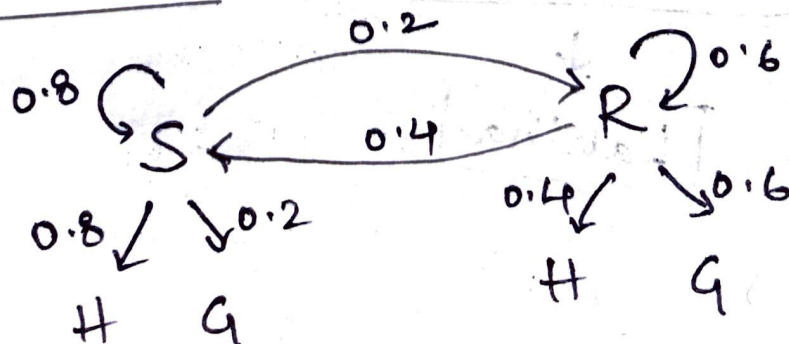
Sunny Rainy S S Rainy S R

Transition Probabilities



Observation: Can observe the mood of Dev } based on
 Weather: Hidden state (Inferred) } observer (lalitha)

Hidden Markov Model



Questions

1. How did we find the probabilities?
2. What's the probability that a random day is S (or) R?
3. If Dev is H today, what's the prob that is S (or) R?
4. If for three days Dev is H, G, H, what was the weather?

Answers:

① Observation

S S S S R R R S S S S R R S S S

Sunny	$S \rightarrow S$	8	0.8
	$S \rightarrow R$	2	0.2
Rainy	$R \rightarrow S$	2	0.4
	$R \rightarrow R$	3	0.6

Transition Probs

S S S S R R R S S S S R R S S S
G H H G G H G H H G H H

$S \rightarrow H$	8	0.8
$S \rightarrow G$	2	0.2
$R \rightarrow H$	2	0.4
$R \rightarrow G$	3	0.6

Emission Probs

②

S	10	$\frac{2}{3}$
R	5	$\frac{1}{3}$

Sunny (or) Rainy

$$S = 0.8S + 0.4R$$

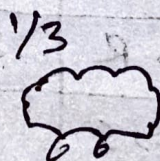
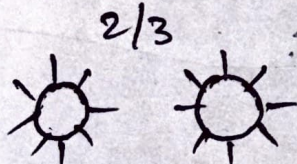
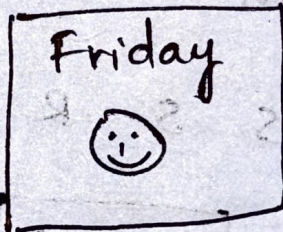
$$S + R = 1$$

$$R = 0.2S + 0.6R$$

③

$P(H/S)$

$P(H/R)$



$$P(S) = \frac{2}{3}$$

$$P(R) = \frac{1}{3}$$

H	H	H
H	H	H
H	H	G
H	H	G

0.8

0.4

$$P(S/H) = \frac{8}{10}$$

$$P(R/H) = \frac{2}{10}$$

$$P(S/G) = \frac{2}{5}$$

$$P(R/G) = \frac{3}{5}$$

0.6

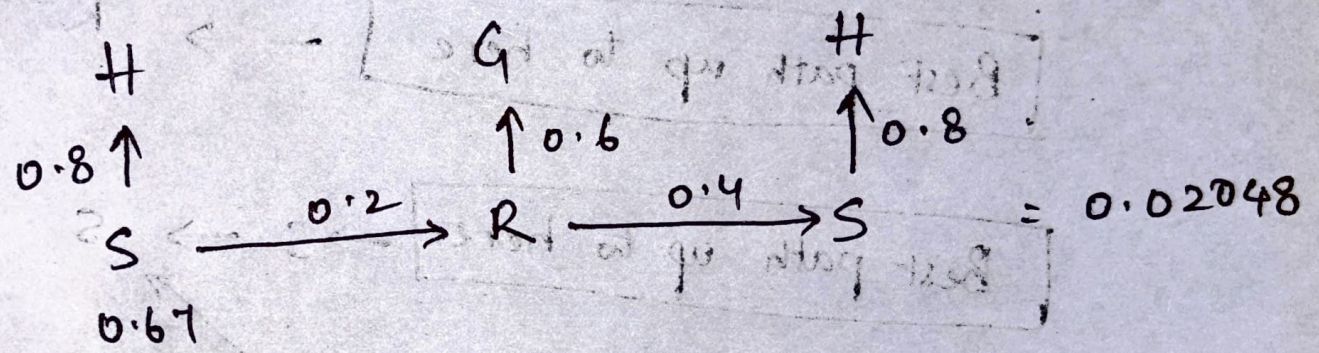
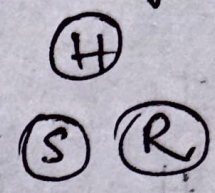
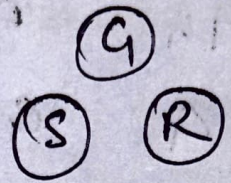
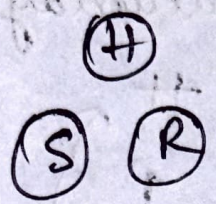
Q

Weather

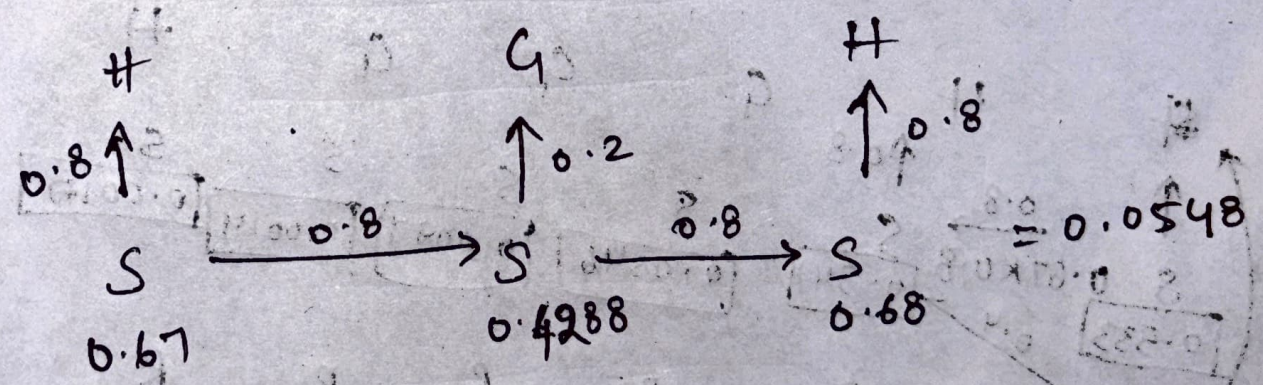
Wednesday

Thursday

Friday

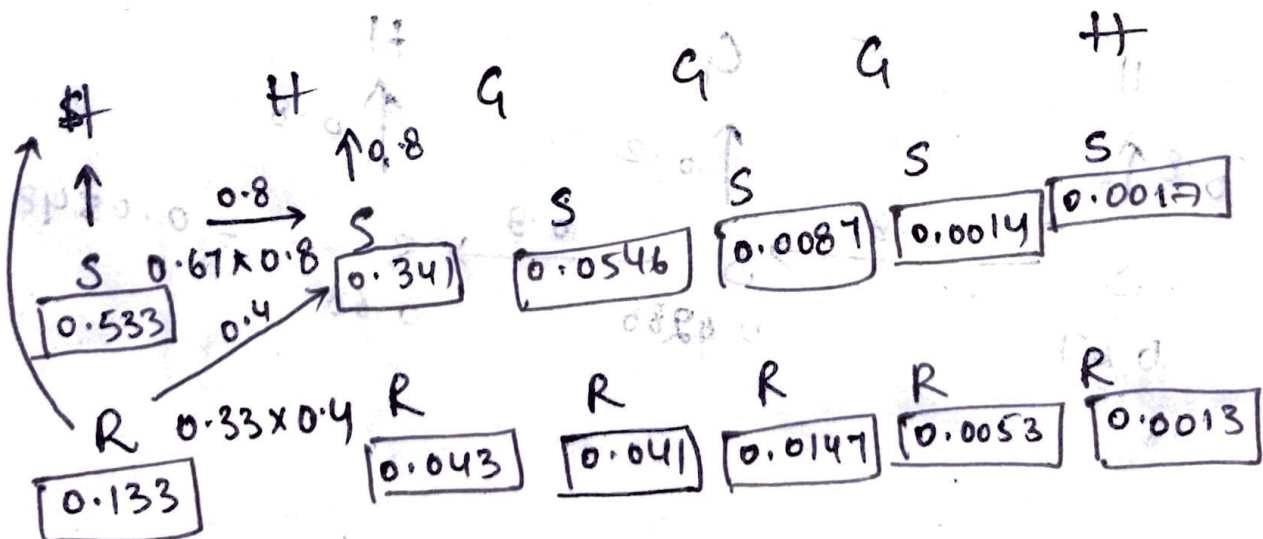
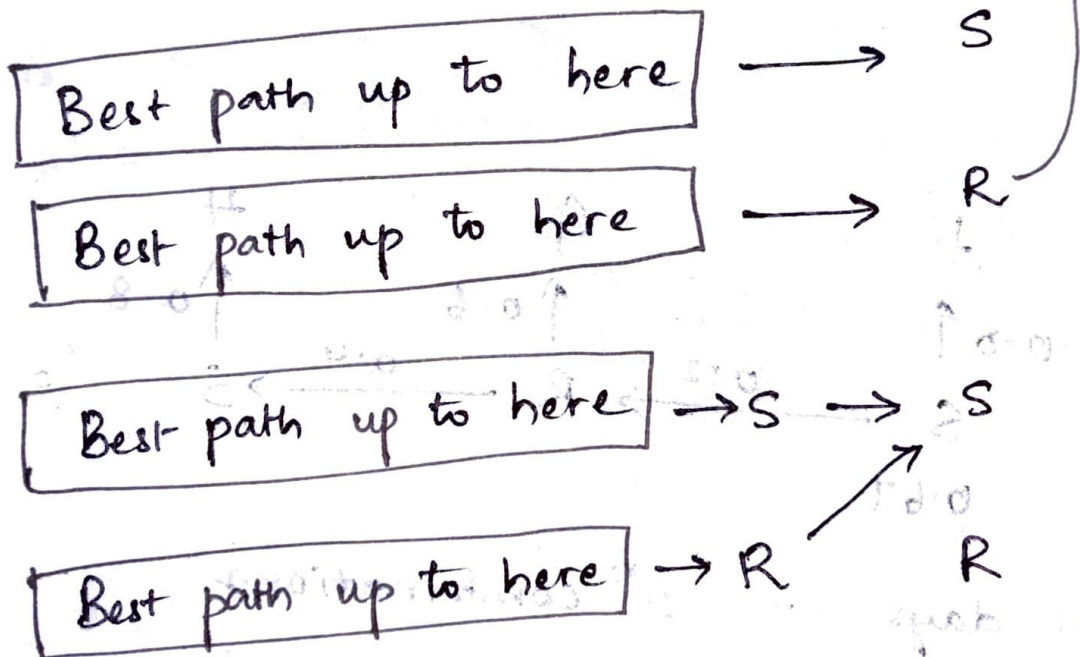


n days $= 2^n$ combinations



Viterbi Algorithm (Dynamic Programming)

Monday T W Th F Saturday
H H G G G H



best Previous prob $\times$ emission $\times$ transition prob

